DIVVY BIKE SHARE
Member vs Casual Rider Behavior Analysis Data Analytics Capstone Case Study | 2023–2025

Tools: Oracle SQL | Microsoft Excel | Tableau

Table of Contents

Contents

1. Introduction 3
2. Data Preparation 5
3. Data Processing 7
4. Data Analysis 10
5. Data Visualization and Communication 13
6. Business Recommendations 16
7. Conclusion 21
8. Data Sources and References 24

1. Introduction

This case study analyzes historical Divvy Bike Share trip data to identify behavioral differences between annual members and casual riders.

Divvy is a bicycle-sharing service in Chicago that provides access to bicycles for transportation and recreation. Riders are categorized as either annual members or casual customers.

The business challenge is to understand how these two customer groups use the service differently and identify opportunities to encourage casual riders to purchase annual memberships.

Business Problem

Annual memberships can provide a recurring revenue stream and support longer-term customer relationships. However, before developing an effective membership marketing strategy, the business needs to understand how casual riders' behavior differs from that of existing members.

This analysis examines riding frequency, trip duration, weekday and weekend usage, hourly riding patterns, bicycle preferences, and popular starting stations.

Business Objective

Identify meaningful differences between annual members and casual riders and recommend data-informed strategies that could increase annual membership conversions.

Key Business Questions

How do annual members and casual riders use Divvy bicycles differently?

What riding patterns suggest opportunities to market annual memberships to casual riders?

Which marketing strategies should the business prioritize based on the findings?

Project Scope

Analysis period: January 2023–December 2025

Dataset: Publicly available Divvy Bike Share historical trip records

Records analyzed: 17,131,207 cleaned rides within the analysis period

Tools: Oracle SQL Developer, SQL, Microsoft Excel, and Tableau

Deliverable: Interactive Tableau dashboard, documented analysis, and business recommendations

Analytical Approach

The project uses SQL to prepare and aggregate historical trip data and Tableau to visualize differences between rider groups.

The analysis is descriptive. Although riding patterns may suggest commuting or recreational behavior, trip records do not establish individual riders' travel purposes or whether a casual rider would purchase a membership.

Recommendations will therefore be presented as business hypotheses to test rather than guaranteed outcomes.

2. Data Preparation

This analysis uses publicly available historical bicycle trip data published by Divvy Bike Share, Chicago's bicycle-sharing system.

The dataset contains individual bicycle trip records that support comparisons between annual members and casual riders.

Original data source: Divvy Trip Data

Data Collection

Historical trip records were collected from monthly CSV files and imported into an Oracle database for analysis.

The broader collection contained 76 monthly files spanning April 2020 through September 2026, with some months unavailable. The analysis was limited to January 2023 through December 2025 to establish a consistent three-year reporting period.

The imported dataset contained approximately 36.3 million raw records.

Dataset Structure

The trip records contain fields such as:

- Ride ID
- Bicycle type
- Trip start timestamp
- Trip end timestamp
- Starting station name and ID
- Ending station name and ID
- Starting and ending coordinates
- Membership classification (member or casual)

These fields allow the analysis to examine trip duration, riding patterns, bicycle preferences, and station usage.

Data Relevance

The dataset is appropriate for comparing annual members and casual riders because it records membership classification alongside trip characteristics.

Trip timestamps support hourly, daily, and monthly analyses, while bicycle type and station information provide additional insight into how each customer group uses the service.

Data Limitations

Several limitations must be considered when interpreting the results:

- The data represents recorded bicycle trips rather than individual customers. A rider may appear in multiple trip records.
- The dataset does not establish why a trip occurred, so commuting and recreational purposes cannot be confirmed.
- Some records contain missing starting-station names, limiting station-level analysis.
- The data does not contain sufficient information to determine whether individual casual riders later purchased annual memberships.
- The findings describe historical behavior and cannot independently establish the effectiveness of proposed marketing strategies.

These limitations will be considered when interpreting findings and developing business recommendations.

3. Data Processing

The Divvy Bike Share dataset was processed using Oracle SQL Developer and Oracle Database Express Edition (XE).

The objective of this phase was to prepare a consistent, reliable dataset for analyzing differences between annual members and casual riders.

The processing workflow included importing historical trip records, identifying duplicate records, validating trip timestamps, calculating ride durations, and creating a cleaned SQL view for subsequent analysis.

Database Environment

Database: Oracle Database Express Edition (XE)

SQL interface: Oracle SQL Developer

Database schema: TEST_USER1

Cleaned SQL view: DIVVY_TRIPS_CLEAN

A centralized Oracle database was used to manage and analyze the large volume of historical trip records.

Data Cleaning Procedures

The following procedures were applied to prepare the dataset:

1. Consolidation of historical records

Monthly CSV files were imported into the Oracle database to create a consolidated dataset for analysis.

2. Duplicate ride identification

Duplicate records were addressed using ride identifiers to prevent repeated trips from inflating ridership statistics.

3. Trip duration validation

Trip start and end timestamps were used to calculate ride durations. Records with zero or negative ride durations were excluded from the cleaned dataset.

4. Membership classification

The member/casual field was used to distinguish annual members from casual riders for comparative analysis.

5. Date filtering

The analytical reporting period was restricted to January 1, 2023, through December 31, 2025.

6. Missing station information

Records with missing starting-station names were excluded from the starting-station rankings. These records were not automatically excluded from other analyses where station information was unnecessary.

Data Validation and Record Counts

The following record counts were established during data preparation:

Processing stage	Number of records
Raw imported trip records	36,328,528
Cleaned trip records across the imported dataset	35,686,178
Cleaned rides within the 2023–2025 analysis period	17,131,207

The difference between raw and cleaned records reflects the combined effect of the cleaning rules. These counts alone do not establish how many records were removed by each individual rule.

Analytical Dataset

The final reporting dataset contained **17,131,207 cleaned trips** during the three-year analysis period.

The data was aggregated using SQL queries to produce datasets for six Tableau visualizations:

1. Monthly ridership trends
2. Average ride duration
3. Weekday versus weekend usage
4. Hourly riding patterns
5. Bicycle type preferences

6. Top 10 starting stations by rider type

The aggregated query results were exported as CSV files and used as data sources for the interactive Tableau dashboard.

Quality Assurance

Data quality checks focused on record counts, ride-duration validity, membership classifications, and consistency between SQL query results and Tableau visualizations.

Station-level results were interpreted separately because missing starting-station information limits the completeness of those rankings.

This processing workflow established the foundation for comparing rider behavior across the selected reporting period.

4. Data Analysis

The objective of this phase was to identify meaningful differences in how annual members and casual riders use Divvy Bike Share.

Using Oracle SQL, the cleaned trip records were aggregated to examine ridership distribution, trip duration, weekday and weekend behavior, hourly usage patterns, bicycle preferences, and starting-station activity.

The analysis covered 17,131,207 cleaned trips from January 2023 through December 2025.

1. Ridership Distribution

Of the 17,131,207 rides analyzed:

Rider Type	Total Rides	Percentage
Annual Members	10,921,833	63.8%
Casual Riders	6,209,374	36.2%
Total	**17,131,207**	**100%**

Key Finding: Annual members accounted for nearly two-thirds of recorded trips, indicating that members represented the larger share of system usage during the analysis period.

These percentages describe trips, not the number of unique customers.

2. Average Ride Duration

Rider Type	Average Duration
Annual Members	12.55 minutes
Casual Riders	25.35 minutes

Casual rides lasted approximately twice as long as member rides on average.

Key Finding: Casual riders generally take longer trips, suggesting different usage patterns between the two customer groups.

This difference may be consistent with leisure-oriented riding, although trip purpose cannot be established from duration alone.

3. Weekday vs. Weekend Usage

Rider Type	Weekend Share	Weekday Share
Annual Members	23.90%	76.10%
Casual Riders	37.16%	62.84%

Both groups recorded more trips on weekdays than weekends. However, casual riders had a substantially higher proportion of weekend trips.

Key Finding: Casual riders demonstrate a stronger relative preference for weekend riding, while annual members show a greater concentration of weekday activity.

This difference may help identify opportunities for targeted weekend membership promotions.

4. Hourly Riding Patterns

Hourly trip analysis revealed differences in riding activity throughout the day.

Annual members showed pronounced activity around typical morning and evening commuting hours, including approximately 8:00 AM and 5:00 PM.

Casual riding activity was more concentrated during daytime and afternoon hours.

Key Finding: Member riding patterns are consistent with regular transportation use, while casual riding patterns may reflect more flexible travel schedules.

These patterns are descriptive and do not prove that particular trips were taken for commuting or recreation.

5. Monthly Ridership Trends

Monthly ridership analysis was used to compare how member and casual trip volumes changed throughout the year.

Both groups demonstrated seasonal variation, with riding activity generally increasing during warmer months and declining during colder months.

Casual ridership exhibited pronounced seasonal changes.

Key Finding: Seasonality should be considered when planning marketing campaigns, particularly those aimed at casual riders.

6. Bicycle Type Preferences

Bicycle type preferences were analyzed separately for annual members and casual riders.

The analysis compared the distribution of recorded trips across available bicycle categories to identify differences in equipment usage.

Key Finding: Bicycle preferences provide an additional dimension for segmenting riders and developing targeted messaging.

Specific marketing decisions should be based on the observed proportions in the Tableau visualization rather than assuming one bicycle category is universally preferred.

7. Starting-Station Analysis

The ten most frequently used starting stations were identified separately for annual members and casual riders.

Station rankings revealed where each group generated substantial trip activity.

Records with missing starting-station names were excluded from this analysis.

Key Finding: High-activity stations may provide useful locations for geographically targeted membership campaigns.

However, station popularity alone does not establish whether a location would produce high membership conversion rates.

Overall Analytical Findings

The analysis identified several meaningful behavioral differences:

- Annual members accounted for the majority of recorded rides.
- Casual rides were approximately twice as long as member rides.
- Casual riders had a higher proportion of weekend trips.
- Member activity showed pronounced morning and evening peaks.
- Seasonal ridership patterns were especially relevant when considering casual-rider marketing.
- Bicycle preferences and starting-station activity offered additional opportunities for customer segmentation.

Together, these findings establish a foundation for developing and testing targeted strategies to encourage casual riders to consider annual memberships.

5. Data Visualization and Communication

Visualization Strategy

An interactive Tableau dashboard was developed to communicate the behavioral differences between annual members and casual riders.

The dashboard presents six visualizations that allow viewers to compare ridership patterns across multiple dimensions.

The objective was to translate the SQL analysis into a clear, accessible presentation that supports data-informed business decisions.

Dashboard Components

The dashboard includes six visualizations:

1. **Monthly Ridership Trends:** Compares trip volumes and seasonal patterns between members and casual riders.
2. **Average Ride Duration:** Highlights differences in average trip length.
3. **Weekday vs. Weekend Usage:** Compares riding activity across weekdays and weekends.
4. **Hourly Riding Patterns:** Identifies differences in riding activity throughout the day.
5. **Bike Type Preferences:** Examines bicycle usage by membership classification.
6. **Top 10 Starting Stations:** Identifies frequently used starting stations for each rider group.

Interactive Functionality

The dashboard includes interactive filtering to support deeper exploration of rider behavior.

Users can select a Member or Casual data point within the Monthly Ridership Trends visualization to filter the other five charts.

Clearing the selection restores the full comparison.

This functionality allows viewers to focus on one customer segment without navigating between separate dashboards.

Visual Design

Consistent colors were used to distinguish the two rider groups:

- **Blue:** Annual members
- **Orange:** Casual riders

The dashboard also includes descriptive titles, instructions for interactive filtering, and a data limitations statement.

Published Tableau Dashboard

Dashboard Title: Divvy Bike Share | Member vs. Casual Rider Behavior Analysis

Analysis Period: January 2023–December 2025

Interactive Dashboard:

https://public.tableau.com/views/Divvy_Bike_Share_Capstone_v1/DivvyBikeShareMembervsCasualAnalysis

The published dashboard provides an accessible way for recruiters, hiring managers, and other viewers to explore the results.

Communication of Findings

The dashboard emphasizes several findings relevant to the membership-conversion business problem.

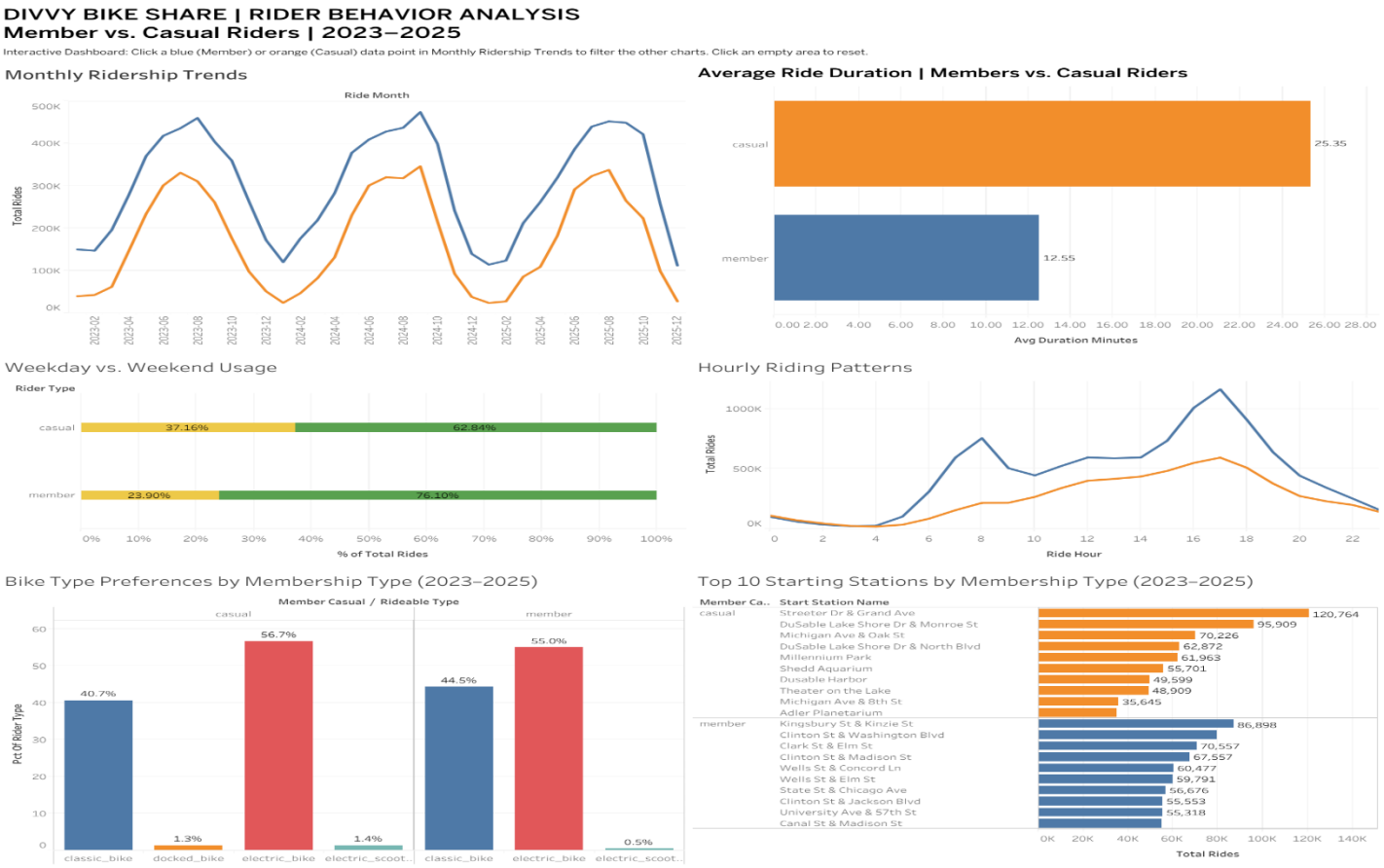
Annual members account for the majority of recorded rides, while casual riders take longer trips and demonstrate a higher proportion of weekend usage.

Member activity is concentrated more strongly around typical commuting hours, while casual activity shows different hourly and seasonal patterns.

These differences provide a foundation for developing targeted marketing hypotheses.

However, the dashboard describes historical trip behavior rather than directly measuring customer motivations or future membership conversions.

Divvy Bike Share — Interactive Rider Behavior Dashboard



[View the Interactive Dashboard on Tableau Public]

6. Business Recommendations

The analysis identified meaningful differences between annual members and casual riders that may help inform membership-conversion strategies.

Casual riders accounted for 36.2% of recorded trips, took longer rides on average, and demonstrated a higher proportion of weekend usage than annual members.

These findings suggest opportunities to develop targeted marketing campaigns rather than relying exclusively on broad, undifferentiated advertising.

The following recommendations are proposed for business consideration and testing.

Recommendation 1: **Target Weekend and Recreational Riding Patterns**

Proposed Strategy

Develop membership marketing campaigns aimed at casual riders during periods of elevated weekend and seasonal activity.

Potential initiatives include weekend-focused promotional messaging, membership comparisons, and targeted campaigns during high-ridership months.

Supporting Evidence

- Casual riders recorded 37.16% of their trips on weekends, compared with 23.90% for annual members.
- Casual trips averaged 25.35 minutes, compared with 12.55 minutes for members.
- Monthly ridership patterns demonstrated substantial seasonal variation.

Advantages

- Aligns marketing efforts with observed casual-rider activity.
- Provides opportunities to test campaigns during periods of higher demand.
- Allows marketing messages to emphasize the potential value of membership for repeat users.

Disadvantages

- Weekend usage does not necessarily indicate a desire for annual membership.
- Seasonal riders may not find annual membership economically attractive.

- Promotional discounts could reduce revenue without generating enough additional conversions.

Recommendation 2: **Introduce Membership Promotions at High-Activity Stations**

Proposed Strategy

Test location-based membership promotions at starting stations with high casual-rider trip volumes.

Promotions could include QR codes linking to membership information, station-specific messaging, or limited pilot campaigns.

Supporting Evidence

The starting-station analysis identified locations with substantial casual-rider activity.

These locations may provide opportunities to reach riders when they are actively using the service.

Advantages

- Focuses marketing resources on locations with demonstrated trip activity.
- Enables comparison of campaign performance across participating stations.
- Can be implemented initially as a limited pilot.

Disadvantages

- High trip volume does not guarantee high membership-conversion potential.
- Missing station information limits the completeness of location-based findings.
- Physical promotional materials may require additional coordination and maintenance.

Recommendation 3: **Test Membership Value Messaging for Frequent Casual Riders**

Proposed Strategy

Develop marketing messages that explain the potential benefits of annual membership for customers who ride repeatedly.

Where permitted and supported by available customer-account data, the business could test personalized messages based on riding frequency.

Supporting Evidence

Casual riders contributed more than 6.2 million trips during the analysis period.

However, the trip-level dataset does not identify how many unique casual customers generated those trips or how frequently an individual customer rode.

Advantages

- Connects membership messaging to potential customer value.
- Allows testing of different promotional offers and messages.
- Could support more relevant marketing if customer-level data becomes available.

Disadvantages

- The analyzed dataset cannot identify frequent individual casual riders.
- Personalized campaigns would require additional customer-account data and appropriate privacy safeguards.
- Membership benefits and pricing would need to be evaluated before making savings claims.

Recommended Strategy

Prioritize Recommendation 1: Target Weekend and Seasonal Casual Riding Patterns.

This strategy is the strongest initial choice because it is directly supported by the observed differences in weekend usage, ride duration, and seasonal activity.

A limited marketing pilot would allow the business to evaluate whether targeted messaging increases membership conversions before committing to a larger campaign.

Station-based promotions could be tested as a secondary initiative, while personalized frequency-based marketing should be considered only when suitable customer-level data becomes available.

All proposed strategies should be evaluated through controlled testing rather than assuming that historical riding patterns will automatically translate into membership purchases.

Implementation Plan

To evaluate the recommended strategy, Divvy could conduct a 90-day pilot campaign focused on casual riders during weekends and periods of high seasonal demand.

The pilot should test whether targeted membership messaging produces measurable improvements in annual membership conversions.

The following timeline is a proposed plan, not an initiative that has already been implemented.

Timeline	Proposed Action	Responsible Team
Days 1–15	Review membership pricing, identify target periods, and establish baseline conversion metrics	Marketing and Business Analytics
Days 16–30	Develop promotional messaging, select campaign channels, and design a controlled experiment	Marketing and Customer Experience
Days 31–75	Launch the pilot campaign and monitor performance	Marketing and Business Analytics
Days 76–90	Evaluate results, estimate financial impact, and recommend whether to expand, revise, or discontinue the campaign	Business Analytics and Finance

Key Performance Indicators (KPIs)

Campaign performance should be evaluated using measurable business outcomes.

1. Membership Conversion Rate

The percentage of eligible casual customers who purchase an annual membership during the measurement period.

Conversion Rate = (New Annual Members / Eligible Casual Customers) × 100

2. Incremental Membership Conversions

The additional memberships attributable to the campaign, estimated by comparing the treatment group with an appropriate control group.

3. Customer Acquisition Cost

The average campaign expenditure required to acquire a new annual member.

Customer Acquisition Cost = Campaign Cost / New Members Attributable to the Campaign

4. Campaign Engagement Rate

The percentage of targeted customers who interact with the promotional campaign.

5. Membership Retention

The proportion of newly acquired members who maintain or renew their membership over a defined period.

Retention would require follow-up beyond the initial 90-day pilot.

Measurement and Evaluation

A controlled experiment is recommended to distinguish campaign effects from seasonal changes in ridership.

Where feasible, comparable eligible customers should be assigned to treatment and control groups. Both groups should be measured over the same period.

The business should compare membership conversion rates, incremental conversions, acquisition costs, and estimated financial returns.

The historical Divvy trip dataset does not contain the customer-level conversion and campaign-cost information needed to calculate these KPIs. Additional internal business data would be required.

Decision Criteria

At the end of the pilot, the business should evaluate whether the campaign:

- Generated a measurable increase in annual membership conversions relative to the control group.
- Acquired members at an economically acceptable cost.
- Produced results that justify further investment.
- Identified customer segments or marketing messages worth testing again.

If the pilot demonstrates positive incremental results and acceptable economics, Divvy could consider expanding the campaign.

If results are inconclusive or unfavorable, the marketing strategy should be revised before additional investment.

7. Conclusion

This case study examined 17,131,207 cleaned Divvy Bike Share trips recorded between January 2023 and December 2025 to identify differences in riding behavior between annual members and casual riders.

Using Oracle SQL for data preparation and analysis, and Tableau for visualization, the project identified several findings relevant to the business objective of increasing annual memberships.

Key Findings

- **Ridership distribution:** Annual members accounted for 63.8% of recorded trips, compared with 36.2% for casual riders.
- **Ride duration:** Casual trips averaged 25.35 minutes, approximately twice the 12.55-minute average for member trips.
- **Weekend usage:** Casual riders recorded 37.16% of their trips on weekends, compared with 23.90% for annual members.
- **Hourly patterns:** Member activity showed pronounced morning and evening peaks consistent with regular transportation patterns.
- **Seasonality:** Monthly ridership varied throughout the year, providing opportunities to consider seasonal marketing strategies.
- **Station activity:** Popular starting stations offered potential locations for targeted membership promotions, although missing station information limited the analysis.

These findings demonstrate that annual members and casual riders exhibit different patterns of bicycle usage.

However, the historical trip data does not identify individual customers, establish trip purposes, or measure membership conversions. Therefore, the analysis supports the development of marketing hypotheses rather than proving which strategies will increase membership sales.

Recommended Business Action

The recommended initial strategy is a controlled marketing pilot focused on weekend and seasonal casual-rider activity.

This approach is supported by the observed differences in weekend usage, average ride duration, and seasonal ridership.

A 90-day pilot would allow the business to test targeted membership messaging, evaluate incremental conversions, and determine whether the strategy warrants broader investment.

Final Next Steps

The following actions are recommended:

1. **Marketing and Business Analytics:** Establish baseline membership conversion metrics and identify suitable campaign periods.
2. **Marketing and Customer Experience:** Develop targeted messaging and implement a controlled pilot.
3. **Business Analytics:** Measure conversion rates, incremental memberships, campaign engagement, and customer acquisition costs.
4. **Finance:** Evaluate campaign economics and the potential return on investment.
5. **Business Leadership:** Review pilot results and decide whether to expand, modify, or discontinue the campaign.

Project Outcomes and Skills Demonstrated

This project demonstrates the application of an end-to-end data analytics workflow to a realistic business problem.

The work included:

- Importing and processing a large historical dataset in Oracle Database.
- Applying SQL-based cleaning, validation, aggregation, and analytical queries.
- Examining more than 17 million cleaned trip records within the selected reporting period.
- Creating six Tableau visualizations and an interactive dashboard.
- Translating analytical findings into actionable business recommendations.
- Identifying data limitations and proposing methods for evaluating future business outcomes.

The completed analysis illustrates how SQL, data visualization, and structured business reasoning can be combined to support decision-making.

Final Reflection

One of the most important lessons from this project is the distinction between identifying behavioral patterns and establishing their underlying causes.

Historical trip data can reveal when, where, and how bicycles are used, but additional customer-level and experimental data would be necessary to determine why riders behave differently and whether a marketing campaign changes their purchasing decisions.

By acknowledging these limitations and recommending measurable experiments, the project provides a practical foundation for further business analysis.

8. Data Sources and References

Divvy Bike Share. (n.d.). *Divvy system data*. https://divvybikes.com/system-data

Divvy Bike Share. *Historical trip data, January 2023–December 2025*. Publicly available monthly bicycle trip records used for this analysis.

Tableau Public. *Divvy Bike Share: Member vs. Casual Rider Behavior Analysis*. https://public.tableau.com/views/Divvy_Bike_Share_Capstone_v1/DivvyBikeShareMembervsCasualAnalysis

Data Attribution and Disclaimer

This is an independent educational portfolio project using publicly available Divvy Bike Share trip data.

The analysis and recommendations are the author's own and do not represent official findings or recommendations from Divvy Bike Share.